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EDUCATION

B.Tech in Artificial Intelligence and Machine Learning
SSIPMT, Raipur

June 2026

PUBLICATION

DocuMind: RAG System for Research Paper Q&A [\[View Paper\]](#) 2026
International Journal on Advanced Computer Engineering and Communication Technology, Vol. 15, No. 1, pp. 160–166
Vyas, S., **Singhal, A.**, Sharma, P., Chauhan, R.P.S., & Chandra, A.

SKILLS

AI / ML	LLMs, RAG, Agentic AI, Multi-Agent Systems, Prompt Engineering, NLP, Deep Learning, Computer Vision, YOLOv8, CNN
Frameworks & APIs	LangChain, ChromaDB, HuggingFace, TensorFlow, FastAPI, Flask, Groq API, Tavily API
Cloud & DevOps	AWS (EC2, S3), Docker, GitHub Actions CI/CD, Render
Programming	Python, SQL, Core Java, C
Tools	Git, VS Code, Jupyter Notebook, Google Colab, Streamlit

PROJECTS

Context — Multi-Agent AI News Explainer [\(GitHub\)](#) [\(Live Demo\)](#) Jul 2026 – Present
Python, Flask, Groq API, Tavily API, Multi-Agent Orchestration, Gunicorn, Render

- Architected a 6-agent pipeline (Discovery, Triage, Researcher, Analyst, Editor, Classifier) that autonomously discovers current news, verifies sources across trusted domains, and synthesizes multi-angle explainers with full source attribution.
- Implemented dynamic query planning where a Researcher agent generates topic-specific search angles at runtime, paired with a triage-based routing layer that classifies stories by required depth to optimize API usage.
- Deployed a live, self-refreshing production system on Render with a Flask REST API, on-demand live-search endpoint, and automated batch content generation pipeline.

RAG-Based Knowledge Management System [\(GitHub\)](#) Jan 2026 – Mar 2026
Python, LangChain, ChromaDB, FastAPI, AWS EC2/S3, Docker, HuggingFace, GitHub Actions

- Deployed a production RAG application on Hugging Face Spaces with Docker and CI/CD via GitHub Actions; integrated AWS S3 for secure cloud document storage.
- Built end-to-end PDF-to-Q&A pipeline using LangChain and ChromaDB for semantic retrieval with source-backed responses; research on this system published in IJACECT 2026.
- Implemented semantic search with MMR retrieval to reduce redundant chunks and improve answer relevance across multi-document corpora.

EXPERIENCE

AI/ML Intern Sep 2024 – Dec 2024
FTV Salon Academy *Remote*

- Built and deployed a DialogFlow + FastAPI chatbot that automated customer interactions, cutting average response time by 50% in simulated production workflows.
- Developed a responsive business website (HTML, CSS, Bootstrap) to showcase the chatbot integration, improving usability and client presentation quality.